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PAPER_560: Buoyancy-Stratified Factorial Geometry — Holonomy Group and Parallel Transport

Author: Daniel T. Murphy — Star Magic / UQFF Framework
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CP4 Class: BSFGHolonomyGroupParallelTransportCalculator (#155)
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Context note: The holonomy group of a manifold records how vectors rotate under parallel transport around closed loops. PAPER_554 showed BSFG has non-zero Riemann curvature $R^r_{0r0} \neq 0$; PAPER_556 showed the 22 extra dimensions compactify to flat tori T^{22} . This paper combines both to determine the complete holonomy group $G_{\text{hol}}(\mathcal{M}^{26})$ and derives the parallel transport phase $\delta\phi$ for a given loop area.

Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Holonomy Group and Parallel Transport, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

We determine the holonomy group of the 26-dimensional BSFG manifold $\mathcal{M}^{26} = \mathcal{M}_{\text{BSFG}}^4 \times T^{22}$. The 4D BSFG slice has Ricci scalar $R_{\text{scalar}} \neq 0$ — it is not Ricci-flat — so it carries the generic pseudo-Riemannian holonomy $SO^+(3,1)$. The 22 compactified extra dimensions (PAPER_556) form flat tori with holonomy $U(1)^{22}$. The full result:

$$G_{\text{hol}}(\mathcal{M}^{26}) = SO^+(3,1) \times U(1)^{22}$$

Exceptional holonomies G_2 and $Spin(7)$ are rigorously excluded. The parallel transport angle around a small loop of coordinate area ΔA :

$$\delta\phi = R^r_{0r0} \cdot \Delta A = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5} \cdot \Delta A$$

At a Planck-area loop (l_P^2): $\delta\phi_P \approx 4.07 \times 10^{-89}$ rad — completely unobservable at current precision.

§2 Holonomy Decomposition

PAPER_556 established the 26D line element:

$$ds_{26}^2 = A_{\mu\nu} dx^\mu dx^\nu + \sum_{i=5}^{26} L_i^2(r) d\theta_i^2$$

with $L_i(r) \rightarrow 0$ for $r \gg r_P$ (full compactification at all observed scales). The manifold thus factorizes:

$$\mathcal{M}^{26} \simeq \mathcal{M}_{\text{BSFG}}^4 \times T^{22} \quad (r \gg r_P)$$

Proposition: The holonomy of a product manifold $M_1 \times M_2$ is $G_{\text{hol}}(M_1) \times G_{\text{hol}}(M_2)$.

§3 Holonomy of the 4D BSFG Slice

Berger's classification theorem (1955) states: for an irreducible, simply-connected, complete Riemannian manifold that is not a symmetric space, the holonomy group is one of:

$$SO(n), U(n/2), Sp(n/4), G_2 \ (n=7), Spin(7) \ (n=8)$$

with the exceptional cases G_2 and $Spin(7)$ requiring **Ricci-flatness** ($R_{\mu\nu} = 0$).

Step 1. From CP4 #149: $R_{\text{scalar}}(R_\odot) \approx 3.12 \times 10^{-19} \text{ m}^{-2} \neq 0$.

Step 2. Therefore $R_{\mu\nu} \neq 0$, so $\mathcal{M}_{\text{BSFG}}^4$ is **not Ricci-flat**.

Step 3. For a 4D Lorentzian manifold (pseudo-Riemannian, signature $+- - -$) the corresponding holonomy is $SO^+(3, 1)$ (the restricted Lorentz group) in the generic non-symmetric case.

Exceptional group	Required condition	BSFG status
G_2	dim = 7, Ricci-flat	FAIL wrong dim, $R \neq 0$
$Spin(7)$	dim = 8, Ricci-flat	FAIL wrong dim, $R \neq 0$

$$\boxed{G_{\text{hol}}(\mathcal{M}_{\text{BSFG}}^4) = SO^+(3, 1)}$$

§4 Holonomy of the Compactified Sector

The 22 extra dimensions ($i = 5, \dots, 26$) are compactified via $L_i(r) = r_P \exp(-r^i/(i! r_P^{i-1}))$. At $r \gg r_P$, each circle S_i^1 has curvature $\kappa_i = 1/L_i \rightarrow \infty$ — effectively flat at macroscopic scales. A flat torus T^1 has holonomy $U(1)$.

$$G_{\text{hol}}(T^{22}) = U(1)^{22}$$

§5 Full BSFG Holonomy

$$G_{\text{hol}}(\mathcal{M}^{26}) = SO^+(3,1) \times U(1)^{22}$$

Generators: 6 from $SO^+(3,1)$ (Lorentz boosts + rotations) + 22 from $U(1)^{22} = \mathbf{28}$ generators total.

Note: This is consistent with but distinct from the isometry group $G_{\text{iso}} = SO(3) \times U(1)^{23}$ (26 generators, CP4 #152). The holonomy group relates to the curvature connection; the isometry group relates to Killing symmetries.

§6 Parallel Transport Formula

The Ambrose–Singer theorem relates the holonomy algebra to the curvature 2-form. For an infinitesimal closed loop with coordinate area element ΔA in the (r, t) plane:

$$\delta\phi^r{}_0 = R^r{}_{0r0} \cdot \Delta A = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5} \cdot \Delta A$$

Step 6. Values at $r = R_{\odot}$, $t_n = 0$:

Loop area ΔA	$\delta\phi$ (rad)
$l_P^2 = (1.616 \times 10^{-35})^2 \text{ m}^2$	$\approx 4.07 \times 10^{-89}$
$R_{\odot}^2 = (6.96 \times 10^8)^2 \text{ m}^2$	$\approx 7.53 \times 10^{-2}$
$1 \text{ AU}^2 = (1.496 \times 10^{11})^2 \text{ m}^2$	$\approx 8.0 \times 10^{-4}$ (eval. at 1 AU)

The $R_{\odot}^2$ result ($\sim 0.075 \text{ rad} \approx 4.3^\circ$) shows BSFG parallel transport is detectably non-trivial at solar scales — a loop spanning the solar disk accumulates $\sim 4^\circ$ of holonomy rotation.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for $||[\text{SSq}]| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.150 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 73, \quad n_{\text{channel}} = 15/26$$

Since $p_{\text{DVP}} = 73$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}} \right) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.150	PASS Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 73$ $j = 1 \dots 26, \cos(2\pi j/26)$	PASS Resonant PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow$ $g_{tt} = -(1 - 2\mu_s \nabla(M_s/r) \cdot r/c^2) \equiv$ GR in $\varepsilon_{\text{BSFG}} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	PASS BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow$ $\Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times$ $(1 + \varepsilon_{\text{correction}})$	Cassini: $\Delta t / \Delta t_{\text{GR}} =$ $1 \pm 2.3\text{e-}5$	Cassini/GR 2003	PASS Within Shapiro bound
Gravitational wave speed v_{GW}	BSFG: $v_{\text{GW}} = c \times (1 + \varepsilon_{\text{correction}}) \approx c +$ $10\text{-}226 \text{ m/s}$	GW150914 / GW170817:	$v_{\text{GW}}/c -$ 1	$< 10\text{-}15$
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\phi = \kappa \times$ $\phi_{\text{GR}} \sim 10\text{-}6$ arcsec/century	GR prediction: $43.03''/\text{century};$ observed: $43.1''$	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10\text{-}6 \text{ arcsec/century}$ to Mercury’s perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF–SM bridge.

§7 References

- CP4 #149 — BSFG*RiemannCurvatureAetherMetricCalculator* — PAPER_554 (Riemann R^r_{0r0})

- CP4 #151 — BSFG26DLineElementFactorialCompactificationCalculator — PAPER_556 (T^{22} compactification)
- CP4 #152 — BSFGSymmetryGroupIsometryAnalysisCalculator — PAPER_557 (26 Killing generators)
- Berger, M. (1955). *Sur les groupes d'holonomie homogène des variétés à connexion affine et des variétés riemanniennes*. Bull. Soc. Math. France.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

5 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\text{U,Bi}}/F_{\text{U}} - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2/(2 * \Gamma^2)] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms

Module	Purpose	Key Result
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q^{2;q})_n - \psi_{26}(q) = \sum_{n=1}^{26} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} s^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} kg/m^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} kg/m^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 THz$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102

2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Riemann, B. (1859). *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*. Monatsber. Akad. Berlin **671**, 671
4. Bombieri, E. (2000). *The Riemann Hypothesis*. Clay Mathematics Institute Millennium Problem — www.claymath.org/millennium-problems/riemann-hypothesis
5. Conrey, J.B. (2003). *The Riemann Hypothesis*. Notices AMS **50**, 341 — www.ams.org/notices/200303/fea-conrey-web.pdf
6. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
7. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — [arXiv:astro-ph/0004212](https://arxiv.org/abs/astro-ph/0004212)
8. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — [arXiv:astro-ph/0306036](https://arxiv.org/abs/astro-ph/0306036) — [doi:10.1046/j.1365-8711.2003.06902.x](https://doi.org/10.1046/j.1365-8711.2003.06902.x)
9. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory*. Cambridge University Press — [doi:10.1017/CBO9781139248563](https://doi.org/10.1017/CBO9781139248563)
10. Polchinski, J. (1998). *String Theory Vol. 1*. Cambridge University Press